Evaluating Limits

Section 1: Direct Substitution and Algebraic Techniques

1 Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$.

a. 3/2

b. 0

c. 1

d. Does not exist

2 Evaluate $\lim_{x \to 0} \frac{\sqrt{x+4} - 2}{x}$.

a. 1/4

b. 1/2

c. 0

d. 1/8

3 Evaluate $\lim_{x \to 1} \frac{\frac{1}{x+1} - \frac{1}{2}}{x - 1}$.

a. $-1/4$

b. 1/4

c. $-1/2$

d. 0

Section 2: Trigonometric Limits and the Squeeze Theorem

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4 Evaluate $\lim_{x \to 0} \frac{\sin(5x)}{3x}$.

a. 5/3

b. 3/5

c. 1

d. 0

5 Evaluate $\lim_{x \to 0} \frac{1 - \cos(x)}{x^2}$.

a. 1/2

b. 0

c. 1

d. Does not exist

6 If $1 - \frac{x^2}{6} \le g(x) \le 1$ for all x near 0, find $\lim_{x \to 0} g(x)$.

a. 1

b. 0

c. 5/6

d. Cannot be determined

Section 3: Infinite Limits and Limits at Infinity

7 Evaluate $\lim_{x \to \infty} \frac{4x^3 - 2x + 7}{5x^3 + 3x^2 - 1}$.

a. 4/5

b. 0

c. ∞

d. $7/(-1)$

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8 Evaluate $\lim_{x \to -\infty} \frac{\sqrt{9x^2 + 4}}{2x - 1}$.

a. $-3/2$

b. $3/2$

c. $9/2$

d. $-9/2$

9 Find $\lim_{x \to 2^+} \frac{3x - 1}{x - 2}$ and $\lim_{x \to 2^-} \frac{3x - 1}{x - 2}$.

a. $+\infty$ and $-\infty$

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 5 and 5

Section 4: Piecewise and One-Sided Limits

10 Let $f(x)$ be defined as $x^2 + 2$ for $x < 1$ and $4 - x$ for $x \geq 1$. Evaluate $\lim_{x \to 1} f(x)$.

a. 3

b. 1

c. 2

d. Does not exist

11 Evaluate $\lim_{x \to 0^+} \frac{|x|}{x}$ and $\lim_{x \to 0^-} \frac{|x|}{x}$.

a. 1 and -1

b. 1 and 1

c. -1 and -1

d. 0 and 0

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12 Find the value of k such that $\lim_{x \to 2} g(x)$ exists, where $g(x)$ is $kx - 3$ for $x \leq 2$ and $x^2 + 1$ for $x > 2$.

a. 4

b. 2

c. 5

d. 3

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